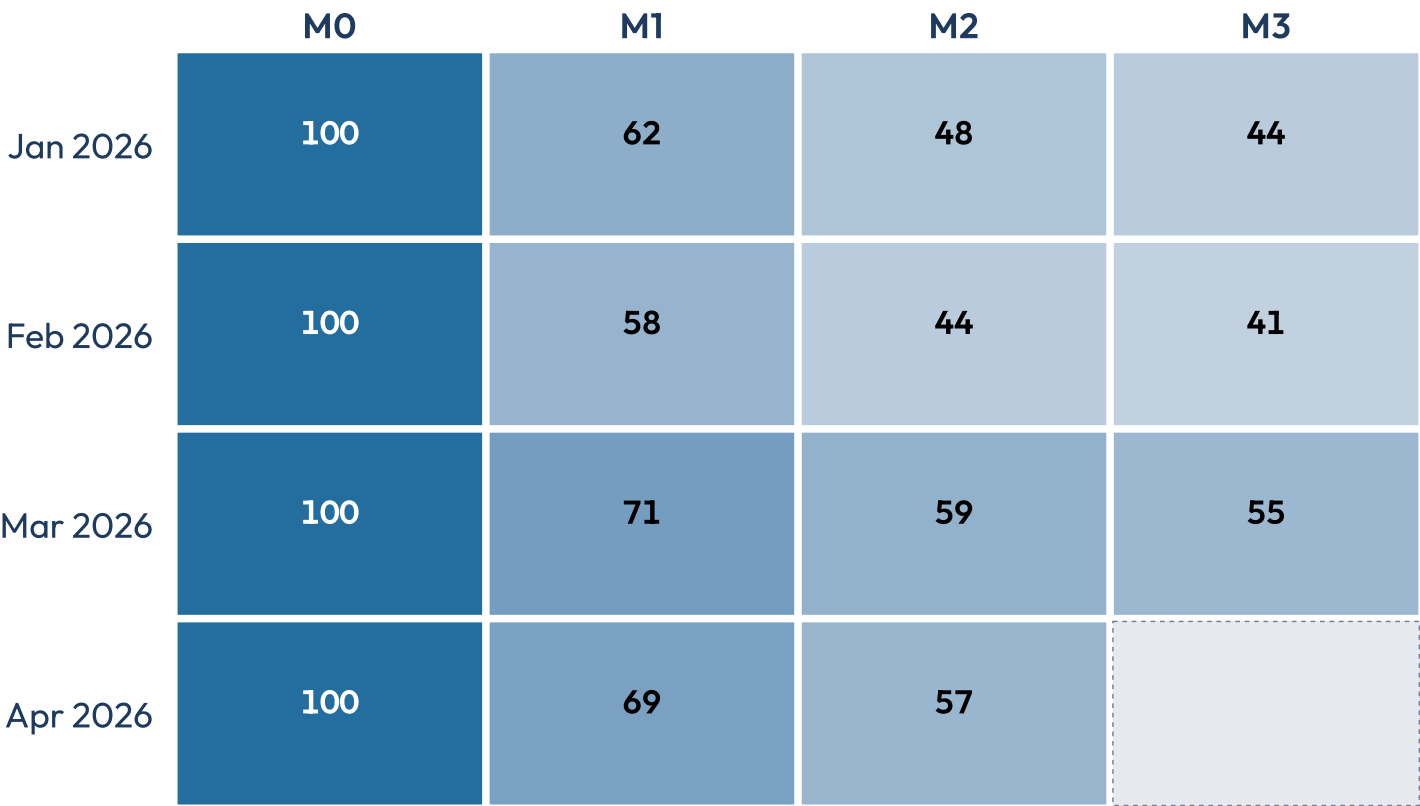


HEATMAP · A NUMERIC MATRIX READ AS INTENSITY

Where does it concentrate?

The chart family could compare magnitudes, decompose a total, plot a trend and score a 2×2 — but it had no way to put a **number at every crossing of two dimensions**. Cohort retention, a risk matrix, usage by hour and weekday: the shape a deck reaches for when the finding is a pattern rather than any single value.

Retention decays fastest in month two.



A missing cell is not a zero.

April has no month-three number because April has not reached month three.

That is the **normal** state of a cohort table, not an authoring error — so the crossing paints the bare neutral with a dotted edge and carries no intensity at all.

Pad it with  instead and the chart states a measurement nobody took, at the bottom of the ramp, where it reads as a collapse.

The ramp is the choropleth's, not a new one.

A heatmap and a `map` encode the same thing — one continuous value as one intensity — differing only in whether the cells are a grid or a geography. So this takes the mechanism `map` already proved: one hue mixed into a **neutral anchor**, never into the page. Mixing toward the canvas looks identical on light and inverts on dark, where a low value sinks *below* the empty cell it is supposed to sit above.

The value's ink flips along the ramp, not with the page.

This is the one place in the family where text sits on a colored mark. A faint cell needs dark ink and a saturated one needs light — on **both** canvases, in opposite directions. The obvious construction, blending the two inks in proportion to the fill, is smooth and unreadable in the middle: it computes mid-grey exactly where the cell is mid-toned. Measured, that rendered 71 at **2.87:1**. The ink steps instead, at a crossover read off the painted pixels rather than guessed — 5.52:1 at its worst now.

RETENTION · THE CEILING

Ten cohorts by twelve months.

	M0	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11
January 2026	100	71	59	55	52	50	48	47	46	45	44	43
February 2026	100	58	44	41	39	38	37	36	35	34		
March 2026	100	62	48	45	43	42	41					
April 2026	100	69	57	54	51							
May 2026	100	66	54									
June 2026	100	64										

When not to reach for it.

Intensity is read approximately — that is its strength for a pattern and its weakness for a comparison. One row of numbers is not a matrix; it is a `bar`, where length is judged precisely. Cells that are verbs or owners belong in `matrix-grid`. A value per country belongs on `map`. And if the decision turns on which of two similar cells is larger, the value belongs on an axis.